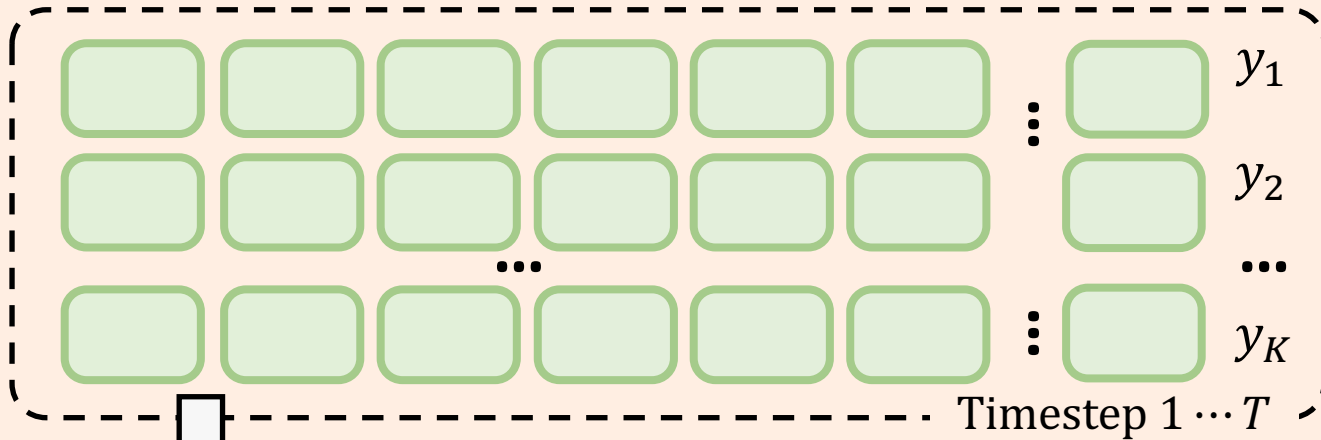


VR-GRPO: Visual Reasoning GRPO



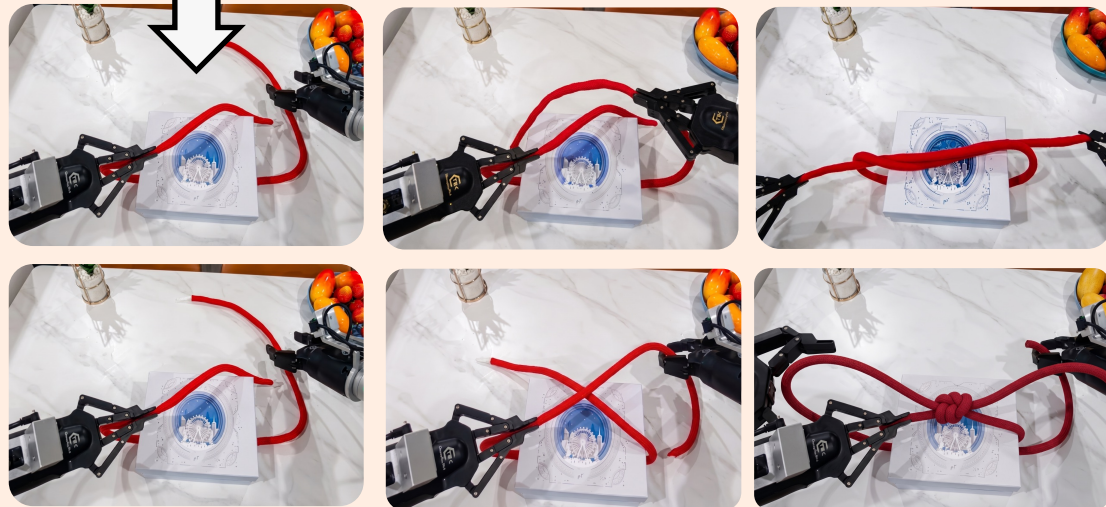
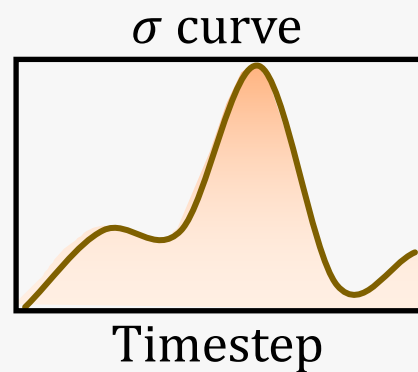
Sub-Step Sampling

$$Z_k(t) \in \mathbb{R}^d, \bar{Z}(t) = E_k[Z_k(t)]$$

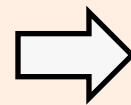
$$\sigma(t) = \sqrt{\frac{1}{K} \sum \|Z_k(t) - \bar{Z}(t)\|_2^2}$$

$$T_{start} = \operatorname{argmax} \sigma(t) - W/2$$

$$T_{end} = \operatorname{argmax} \sigma(t) + W/2$$



Pairwise Groups



Global-Level Assessment

Please evaluate the overall quality of the image sequence:

1. Visual coherence ...
2. Task completion ...

Step-Level Assessment

You will be shown **two clips**, Please compare their quality:

1. Whether the trajectory progresses toward the goal state ...
2. Whether the reasoning follows physical laws ...
3.

Please compare which one is better, **return A or B**. If they are the same, **return TIE**.

Global Reward

R_g^1

R_g^2

R_g^3

...

R_g^K

VLM

Step Reward

R_s^1

R_s^2

R_s^3

...

R_s^K

$$R = R_g - \lambda \cdot |R_g - R_s|$$

Failure Case of Global Quality Assessment



VLM

This sample has smooth actions and reach the final state. Therefore, **its quality is good**.

